

IoT-Enabled Cloud-Based Power Management System Using AWS and PyQt5 Interface

Jency Rufina A
School of Electrical Engineering
Vellore Institute of Technology
Chennai, India

jencyrufina.a2023@vitstudent.ac.in

A Aneeka Reemaz
School of Electrical Engineering
Vellore Institute of Technology
Chennai, India

aneekareemaz.a2023a@vitstudent.ac.in

Vaanmathi R S
School of Electrical Engineering
Vellore Institute of Technology
Chennai, India

vaanmathi.rs2023@vitstudent.ac.in

Gnana Swathika O V
Centre for Smart Grid Technologies
Vellore Institute of Technology
Chennai, India
gnanaswathika.ov@vit.ac.in

Abstract— The need for effective and intelligent methods of power monitoring is high, considering the increased demand for electrical energy caused by the rise in the use of automated systems across multiple platforms. The need for manual monitoring or the dependency on hardware-based equipment, which do not have many facilities aside from manual implementation, makes traditional power monitoring techniques the state of the art in many cases. In the following work, a desktop application based on PyQt5 along with the Amazon Web Services (AWS) platform is used to design an IoT-based power management system, where the MQTT protocol is used for the safe transfer of electrical values from IoT-enabling devices. Serverless data processing and threshold checking are handled by AWS Lambda, and AWS IoT Core manages the connection of devices. AWS RDS is employed for storing power consumption data, and AWS Simple Email Service (SES) is employed for automatically sending notifications whenever abnormal power consumption is detected. Power consumption and system status are displayed in real-time using a graphical user interface developed using PyQt5. Low-latency communication, reliable notification delivery, and a scalable and cost-effective solution for real-time power monitoring in domestic and small-scale industrial applications are demonstrated by experimental results.

Keywords— *PyQt5, Cloud Computing, Power Management, Amazon Web Services (AWS), Internet of Things (IoT), and Energy Monitoring*

I. INTRODUCTION

One of the most vital resources that are currently keeping the modern world ticking is electrical energy, the demand for which is growing rapidly with the advent of industrialization, urbanization, and the increasing use of automation technology. Smart devices, integrated systems, and electrically driven devices are increasingly being used in residential, commercial, and industrial setups. However, due to the increasing demand for energy, the existing power infrastructure is under immense pressure, and hence, efficient electrical energy management is the need of the hour.

Manual meter readings, periodic inspections, or locally installed hardware-based monitoring units are the cornerstones of conventional power monitoring and management solutions. These conventional solutions have several disadvantages, including limited remote accessibility, delayed fault detection, unavailability of real-time information, and poor scalability, despite the fact that they

may provide basic consumption information. In addition to being susceptible to human error, manual data acquisition methods are incapable of detecting sudden anomalies, such as overloads, power theft, or unexpected consumption peaks. Moreover, conventional solutions often lack intelligent data analysis capabilities and are stand-alone, which makes it difficult to take proactive steps and manage power consumption effectively.

There has been a growing emphasis on energy conservation, cost optimization, and reduction of carbon footprint, which has led to the spread of smart energy management solutions. Among them, the Internet of Things (IoT) has been recognized as a game-changing technology that makes it possible to continuously sense, communicate, and control physical systems. IoT-based devices with sensors can perform real-time monitoring of electrical variables including voltage, current, power, and energy. These devices allow data gathering at various locations in an electrical system, hence enhancing situational awareness and facilitating smart control actions.

The problem with IoT devices is that they have limited computational capabilities, memory, and storage. It is generally inefficient and often not possible to process vast amounts of data at the device level. The problem is solved by cloud computing that enables elastic computation, data storage, high availability, and accessibility. Thus, any system that monitors the consumption of power can use the benefits of IoT and cloud solutions such as analytics, data storage, and real-time analysis without the IoT devices getting exhausted.

However, IoT by itself has limitations in terms of computational capabilities, memory, and storage. The idea of processing large amounts of data at the device level is inefficient and, in most cases, not feasible. The issue is resolved by cloud computing, which allows for elastic computation, data storage, high availability, and accessibility. With the combination of IoT and cloud solutions, power monitoring systems can benefit from analytics, data storage, and real-time analysis without straining the IoT devices.

Some researchers have looked into the integration of IoT and cloud computing in the context of power and energy systems. The benefits of cloud and IoT technologies in power distribution networks are limited to sustainability, energy efficiency, and visibility, as Rajagopalan et al. [1] showed. Lu et al.'s [2] cloud-based power system simulation platform

demonstrated how cloud-native solutions could be beneficial for power engineering problems, while also addressing the local software scalability issue. Khattar et al. [3] have emphasized the significance of dynamic power management techniques, which are more compatible with cloud-based monitoring systems, in their comprehensive examination of energy-efficient cloud computing strategies.

Further advancements in the energy monitoring process for specific domains have also been discussed. Kumar and Saini [4] proposed an IoT and cloud-enabled hydropower energy monitoring system, which demonstrated the use of cloud analytics for predictive monitoring. Khalil et al. [5] analysed the feasibility of using cloud computing for relay protection in high voltage substations and demonstrated how cloud processing can be centralized for enhanced reliability and reduced maintenance costs. Xu et al. [6] concentrated on the security and privacy aspects of cloud-based frequency control and proposed the application of homomorphic encryption in multi-area power systems.

Nevertheless, a few cases remain that have not been fully dealt with by the existing literature. Most of the present implementations are focused on either web, based dashboards or the proprietary systems that may need constant internet connectivity, compatibility with different browsers, or complicated setup procedures. The field of lightweight, self, contained, and desktop, based visualization systems has attracted less attention, especially when combined with serverless cloud infrastructure. On top of that, the technical specifics of sending the alert messages automatically, figuring out the anomalies by the threshold, or designing the user interface in a very intuitive way are generally neglected or put aside as a secondary issue.

Apart from that, one of the major drawbacks identified in the current systems is the lack of a comprehensive end to end solution that encompasses the implementation of IoT sensing, cloud communication, real-time processing, data storage, notification, and visualization in a single unified platform. Most of the current works are only able to provide theoretical models or simulation-based analyses without proving the feasibility of the system implementation.

To address the above gaps, this paper introduces a cloud, based power management system with IoT capability which leverages Amazon Web Services (AWS) and a desktop application built with PyQt5. The selection of AWS is due to its comprehensive IoT offerings, serverless computing features, scalability, and security framework. Using AWS IoT Core, devices can be securely connected through lightweight communication protocols like MQTT. It also becomes easy with AWS Lambda to carry out event, driven, serverless processing of data at hardly any operational costs. AWS RDS guarantees reliable and scalable data storage while AWS Simple Email Service (SES) enables the sending of automatic alert emails when power consumption patterns that are anomalous are detected.

The desktop GUI, which is based on PyQt5, is a kind of light and responsive standalone visualization platform that displays the power consumption data in the real time, historical trends, and alert notifications. Unlike the web-based dashboards, the desktop application provides better performance, less dependency on browsers, and a better user experience for local monitoring environments such as control rooms, laboratories, and small industrial setups.

The main objective of this paper is to setup, develop, and experimentally verify a fully automated power measurement chain ranging from data acquisition by IoT, enabled sensors for secure data cloud transmission, real-time processing, automatic alert generation, to user-friendly desktop visualization. The proposed work is expected to be scalable, reliable, cost-effective, and easy to deploy while ensuring low latency and high system availability.

The remaining part of this paper is organized as follows: The system methodology, which includes the architecture, communication protocols, cloud services, and GUI design, is described in Section II. The experimental results and performance analysis are presented in Section III. Section IV gives a summary of this paper. Sections V and VI cover the limitations and future scope, respectively.

II. METHODOLOGY

This section of the paper is a detailed description of the system architecture, communication protocols, security features, user interface design, and validation techniques that were employed in the development of the IoT, enabled cloud, based power management system using AWS and PyQt5. The technique involves the use of hardware communication theories and cloud, based data management approaches to ensure scalability, security, and high reliability of real, time energy monitoring and control.

A. System Architecture

The IoT, cloud-enabled, based power management system that is proposed in this paper, is designed with a layered architecture for scalability, reliability, and ease of deployment. The general system architecture is shown in Fig.1. The architecture is divided into three layers: the sensing layer, the cloud processing layer, and the application layer.

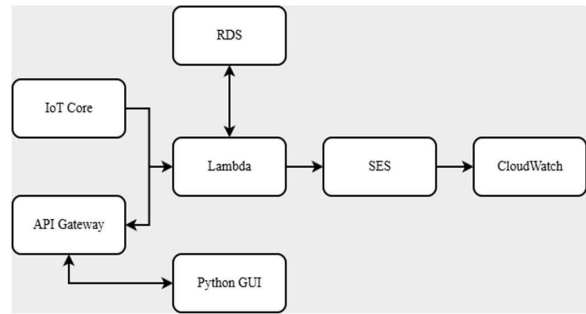


Fig. 1. Block diagram of the designed IoT-enabled cloud-based power management system architecture.

The sensing layer is made up of IoT measurement units with sensing capabilities that are responsible for the measurement of electrical parameters like voltage, current, and power consumption. The loading vibration is measured accurately because these parameters are measured at fixed intervals. The data obtained is pre-processed locally to remove noise and formatting differences before being uploaded to the cloud.

The cloud processing layer is the core part of the system. Using AWS IoT Core as the secure entry point for data ingestion, a communication channel between IoT devices and

cloud services is created and made reliable. Cloud serverless functions are used to further process data, validate readings, and identify abnormal events. The processed data is stored in a central database for future use in historical analysis and reporting.

The application tier consists of a desktop program developed with PyQt5 that enables real-time visualization and interaction. By dividing the system into layers, it is possible to achieve the separation of concerns, which permits each layer to be modified separately without impacting the system.

B. Data Communication and Message Format

Efficient and reliable data transfer are the key to real-time power monitoring. The system employs the MQTT protocol because it is a very light protocol and as such would be appropriate for IoT. Figure 2 illustrates the structure of the MQTT payload sample used for the transmission of the sensor data.

```
Message payload
{
  "device_id": "device_A001",
  "power_usage": 450
}
```

Fig. 2. Example of MQTT message payload showing device ID and power consumption data structure.

Each message comprises sensor readings with time stamps, device information and a checksum field value. MQTT's publish/subscribe communication model supports highly scalable communication. This enables multiple devices to send data at the same time without causing a traffic jam. The system quality of service (QoS) is chosen according to system demands.

C. Cloud-Based Data Processing

Following data collection, AWS IoT Core leverages AWS Lambda functions for serverless computing. The functions perform data validation, threshold comparison, and anomaly detection. The threshold values, which are determined by the acceptable operating ranges, can be dynamically adjusted via the cloud interface.

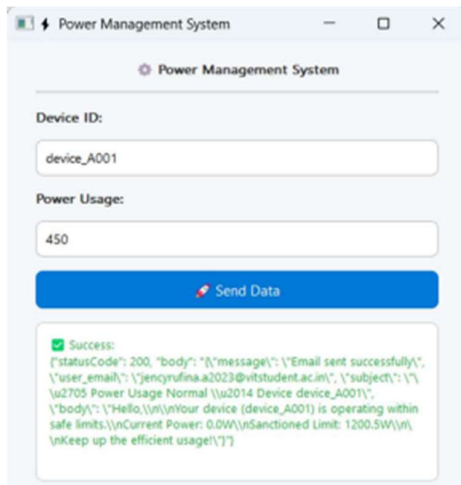


Fig. 3. PyQt5-based GUI application displaying real-time Power usage data and alerts.

An RDS in AWS, which is a cloud computing platform, is storing the validated data in a relational database allowing for fast queries and pattern recognition features. It is also saving a record of the power usage pattern for the last few months. Whenever the system finds power usage anomalies on the customers' side, it issues a warning through AWS SES.

D. PyQt5 Graphical User Interface

The desktop client, implemented in PyQt5, represents our main interface towards displaying and steering the data. Fig. 3 shows a sample screenshot of the GUI with many examples of numerical readouts, power-use curves, and alerts indicators. The GUI is designed to be simple and easy to use, with colour cues and symbolic displays to rapidly discern normal from abnormal states. The continuous retrieval of data from the cloud occurs via RESTful API services exposed by AWS API Gateway. Besides that, robust error-handling routines in the system maintain the stability and operability of the application even with a loss of network connectivity.

III. RESULTS AND DISCUSSION

A. Experimental Setup

For testing the functionality and validation of the system, a controlled environment had to be ensured. A detailed list of the parameters considered during the testing process is provided in Table I. These parameters include, and are not limited to, selecting the sampling interval, communication protocol, and configuration of the cloud server. For testing the parameters and the system as a whole, the experiments were conducted in low and high loading scenarios to ensure complete validation of the system due to different modes of power consumption.

TABLE I
AWS SERVICES AND THEIR ROLES IN SYSTEM IMPLEMENTATION

AWS Service	Role Description	Configuration Highlights
AWS IoT Core	Secure MQTT broker and device management	Device certificates, TLS authentication
AWS Lambda	Serverless event-driven data processing	Configured with 512MB memory, 1min timeout
AWS RDS	User/device data storage and management	PostgreSQL engine, encrypted at rest
AWS SES	Automated email alert delivery	Verified email domains, SMTP integration
AWS API Gateway	RESTful API endpoints for GUI communication	REST methods secured via IAM policies

B. Performance Metrics and Test Results

The numerical results of the performance measurements carried out in the experiments are given in Table II. In this study, we quantitatively investigated the main parameters of data transmission latency, processing time in the cloud, system operating time, and alert response time. The results demonstrate the effectiveness of the proposed system by showing that the low-latency performance of the system is possible for reliable performance.

TABLE II
SYSTEM PERFORMANCE METRICS AND TEST RESULTS

Metric	Value / Range	Description
Lambda Execution Time	~400 ms (average)	Average processing time for serverless functions
Lambda Memory Usage	~100 MB (average)	Memory consumed during function execution
MQTT Message Delivery Latency	< 1 second	Time from device message publish to processing
Email Notification Delay	< 2 seconds	Time taken to send alert emails via AWS SES
MQTT Payload Size	~100 bytes	Average message size sent from IoT devices
System Uptime	99.9%	Measured system availability during tests

C. Discussion

As can be clearly derived from the results, the combination of IoT devices and cloud services plays an important role in the management of the power. Running a serverless setup on the cloud helps a process to occur almost instantaneously. This helps in the early detection of any fault or irregularity. The role of the PyQt5 GUI is to raise awareness and enable users to make smarter decisions through data visualization. The minor delays, attributed to other factors during these busiest periods, seemed to occur sporadically.

During our tests, we were able to simulate IoT devices sending in power usage data to be processed. It averaged about 400ms to process, so we were able to monitor in close to real time. Database queries were consistently under 150ms in every test.

Emails sent using the AWS SES were received within 1-2 seconds as soon as the limits were violated. The PyQt5 application provided near-live data with minimal latency in terms of response.

During stress testing, there were occasional notifications of delays in the delivery of MQTT messages when the message rate was extremely high. These were fixed by tuning the Lambda memory allocation and the QoS settings. Overall, the application has managed an approximately 99.9% availability rate, mostly thanks to the robustness of the AWS services that we use.

IV. CONCLUSION

Overall, the IoT-enabled, cloud-backed power management system represents a case where the sensing device, the cloud service, and the desktop visualization tool can work together. Moreover, the IoT device utilizes the AWS service for cloud power and the PyQt5 library for its desktop visualization tool. Consequently, it provides the real-time viewing, data logging, and alerting of electricity consumption. It has the advantage of scalability and reduced maintenance due to its modular and serverless structure, and it also has good performance as validated by the test results.

V. LIMITATIONS

Like all systems, it has its own advantages and disadvantages.

This system depends on the connectivity of the network. In the event that there is no connectivity to the internet, it will affect the updating of the data.

Currently, the focus is on the monitoring and alerting aspects. There are no built-in features with regards to prediction, fault detection, and self-diagnosis. The system has only been demonstrated within a controlled environment, and going live may cause issues with regards to synchronization, data traffic, and overall reliability.

VI. FUTURE SCOPE

Future advancements may even remove these limitations, thus running the system at its maximum potential. For example, machine learning capability added to the cloud processing layer can facilitate predictive maintenance, anomaly detection, and load forecasting. A feature that would be added would enhance the desktop app and the accessibility of the program, as it would allow the user to work on the move. Furthermore, the system may have evolved to be multi-node, which will be more appropriate for addressing the needs of the industries and smart grids. There exists the possibility of enhancing data integrity and security through a combination of the use of blockchain and green energy resources.

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